

Handling Missing Values

Aim:

To handle missing and invalid values in a hotel dataset to ensure data integrity and prepare the data for analysis and visualization.

Procedure:

1. Import the required libraries pandas and numpy for data manipulation.
2. Load the dataset using `pd.read_csv()` to create a DataFrame.
3. Display the dataset using `df.head()` and check for duplicates using `df.drop_duplicates()`.
4. Use `df.info()` to view data types and non-null counts.
5. Identify invalid or negative numerical values that must be replaced with NaN.
6. Replace invalid values in columns like CustomerID, Bill, and EstimatedSalary using `np.nan`.
7. Correct NoOfPax column values by replacing numbers outside the valid range with NaN.
8. Use `df.loc` for safe assignment to avoid SettingWithCopy warnings.
9. Check for missing data using `df.isnull().sum()`.
10. Impute or drop missing values as necessary based on data requirements.
11. Validate the final dataset to ensure no missing or invalid entries remain.
12. Display the cleaned DataFrame using `df.head()`.

Program:

```
import pandas as pd
import numpy as np

df = pd.read_csv('/content/Hotel_Dataset - Hotel_Dataset (1).csv')
df.head()

df.drop_duplicates(inplace=True)
df.info()

index = np.array(list(range(1, len(df) + 1)))
df.set_index(index, inplace=True)
df

df.loc[df['CustomerID'] < 0, 'CustomerID'] = np.nan
df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan

df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan
df
```

Output:

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	
1	1.0	20-25	4	Ibis	veg	1300.0	2.0	4000
2	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3.0	5900
3	3.0	25-30	6	RedFox	Veg	1322.0	2.0	3000
4	4.0	20-25	-1	LemonTree	Veg	1234.0	2.0	12000
5	5.0	35+	3	Ibis	Vegetarian	989.0	2.0	4500
6	6.0	35+	3	Ibys	Non-Veg	1909.0	2.0	12200
7	7.0	35+	4	RedFox	Vegetarian	1000.0	NaN	21000
8	8.0	20-25	7	LemonTree	Veg	2999.0	NaN	34560
9	9.0	25-30	2	Ibis	Non-Veg	3456.0	3.0	87700
10	10.0	30-35	5	RedFox	non-Veg	NaN	4.0	87700

Result:

Hence, a Python program was successfully written to handle missing and invalid values in the dataset, ensuring the data is clean and ready for further analysis.